

Zekai Zheng

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Biography

I am currently a PhD student in Physics at National University of Singapore (NUS) who is passionate in Biophysics and Astronomy. Beyond academia, I actively engage with the creative arts through film production, photography and music, while also fulfilling my love for traveling and sports like basketball, football, hiking and motorsports.

Education

National University of Singapore

Ph.D. in Physics

Aug 2024 - Present

Singapore

Zhejiang University

B.S. in Physics

Sept 2020 - June 2024

China

- GPA: 3.85/4.00

Academic Publications

- [1] Lin, L., Liu, B. and **Zheng, Z.**, 2025. Resonance capture and stability analysis for planet pairs under Type I disk migration. *Astronomy & Astrophysics*, 702, p.A161.
- [2] Yang, S., Wu, L., **Zheng, Z.**, Ogihara, M., Guo, K., Ouyang, W. and He, Y., 2023. The stability of unevenly spaced planetary systems. *Icarus*, 406, p.115757.

Research Experience

Artificial molecular motor and DNA nanotechnology

(National University of Singapore)

Aug 2024 - Present

- Design Synthesis: Designed complex DNA origami structures and executed high-precision synthesis via sequence-specific thermal annealing protocols.
- System Optimization: Optimized single-stranded DNA molecular motors by applying principles of DNA-RNA hybridization and statistical physics.
- Nanoscale Characterization: Validated structural integrity and monitored real-time molecular binding events using Atomic Force Microscopy (AFM) and Fluorescence Spectroscopy.
- Computational Toolkit: Programming in Python, Molecular simulation with oxDNA package, Cadnano, Origin, AlphaFold3.
- Experimental Toolkit: Atomic Force Microscopy, Fluorescence Spectroscopy (FRET), Gel Electrophoresis (PAGE/agarose), Thermal Cyclers, and NanoDrop Spectrophotometry.

Planet migration and resonance

(Zhejiang University)

Sept 2022 - June 2024

- Theoretical Modeling & Numerical Validation: Developed a unified analytical framework and executed extensive N-body simulations to evaluate the resonance capture and long-term dynamical stability of two-planet systems undergoing convergent Type I migration.
- Analytical Derivations & Key Findings: Derived general stability criteria for first-order resonances across arbitrary mass ratios to classify orbital evolution into different patterns.
- Computational Toolkit: Programming in Python, N-body numerical simulation with REBOUND and REBOUNDx package, Mathematica.

Conferences & Activities

- Poster session at Foundations of Nanoscience (FNANO) 23th Annual Conference, 2026
- Sharing session at NUS Physics Summer Camp, 2025